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TABLE I - CAPACITIES FOR REFRIGERANT -12

MS PART NO.	CAPACITY - BTU/HR: 135°F CONDENSING, 125° VAPOR FREE LIQUID ENTERING TEV											
	-10°F EVAP TEMP 175 PSI PD ACROSS TEV		25°F EVAP TEMP 145 PSI PD ACROSS TEV		-10°F EVAP TEMP 145 PSI PD ACROSS TEV		25°F EVAP TEMP 125 PSI PD ACROSS TEV		50°F EVAP TEMP 100 PSI PD ACROSS TEV		105° COND TEMP 95°F VAPOR FREE LIQUID ENT TEV 50°F EVAP TEMP 35 PSI PD ACROSS TEV	
	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
MS -1	2000	3700	3400	7000								
MS -2	3600	4200	6300	8800								
MS -3					4500	8700	6200	15000				
MS -4					8500	19200	18000	30000				
MS -5					13000	28000	24000	43000				
MS -6									13000	18000	10000	13000
MS -7									28000	34000	19000	22000
MS -8									40000	52000	29000	35000
MS -9									57000	75000	36000	54000
MS -10									72000	78000	49000	54000
MS -11									109000	130000	71000	83000
MS -12									145000	172000	100000	117000

THERMOSTATIC EXPANSION VALVE = T.E.V
PRESSURE DROP = P.D

CAPACITIES: MINIMUM CAPACITY REQUIREMENTS MUST BE MET. MAXIMUM CAPACITIES ARE MERELY A GUIDE FOR THE APPLICATION ENGINEER. THE SUFFIX "A" IN TABLE II OR III DOES NOT AFFECT CAPACITY.

TABLE II - CAPACITIES FOR REFRIGERANT - 22

MS PART NO.	CAPACITY - BTU/HR: 135°F CONDENSING, 125°F VAPOR FREE LIQUID ENTERING T.E.V.											
	-10°F EVAP TEMP 250 PSI P.D.		25°F EVAP TEMP 220 PSI P.D.		-10°F EVAP TEMP 220 PSI P.D.		25°F EVAP TEMP 200 PSI P.D.		50°F EVAP TEMP 175 PSI P.D.		105°F COND TEMP 95°F VAPOR FREE LIQUID ENT T.E.V. 50°F EVAP TEMP 75 PSI P.D.	
	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
MS -13	3250	6020	5530	11380								
MS -14	5850	6830	10250	14300								
MS -15					7550	14600	10350	25000				
MS -16					14250	32200	30000	49800				
MS -17					21800	47000	40000	71500				
MS -18									22200	30700	18880	24600
MS -19									47100	58000	35800	41500
MS -20									68000	88500	54700	66000
MS -21									97000	128000	68000	102000
MS -22									122500	133000	92500	102000
MS -23									185700	222000	134000	156800
MS -24									247000	293000	188800	221000

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This military standard has been approved by the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Review Activity: Army-MR
User Activity: Navy-SH, MC

©ENTIRE STANDARD REVISED

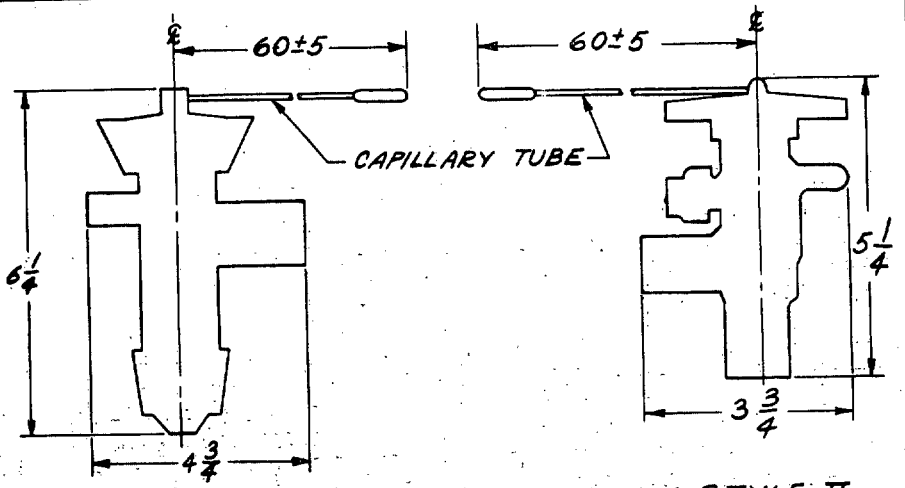
P.A. ARMY — GL Other Cust NAVY — YD USAF — 82	TITLE VALVES, EXPANSION, THERMOSTATIC, REFRIGERANT-12 AND REFRIGERANT-22	MILITARY STANDARD MS 17982
PROCUREMENT SPECIFICATION MIL-V-23450	SUPERSEDES:	SHEET 1 OF 3

DD FORM 1 SEP 67 672-1

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VALVE MAXIMUM OVERALL DIMENSIONS

	MS PART NO. STYLE I	MS PART NO. STYLE II	EQUALIZER TYPE	EQUALIZER CONNECTION	PRESSURE LIMIT MEANS	CONNECTIONS SIZE (IN) INLET	CONNECTIONS SIZE (IN) OUTLET	CONNECTIONS TYPE
TABLE III REFRIGERANT-12	-1	-1A	INTERNAL		YES	3/8	1/2	FLARE
	-2	-2A						
	-3	-3A				1/2	5/8	
	-4	-4A						
	-5	-5A						
	-6	-6A						
	-7	-7A	EXTERNAL	FLARE 1/4	NO			
	-8	-8A				5/8	7/8	SOLDER
	-9	-9A						
	-10	-10A						
	-11	-11A				7/8	1 1/8	
	-12	-12A						
TABLE II REFRIGERANT-22	-13	-13A	INTERNAL		NO	3/8	1/2	FLARE
	-14	-14A						
	-15	-15A				1/2	5/8	
	-16	-16A						
	-17	-17A						
	-18	-18A						
	-19	-19A	EXTERNAL	FLARE 1/4				
	-20	-20A				5/8	7/8	SOLDER
	-21	-21A						
	-22	-22A						
	-23	-23A				7/8	1 1/8	
	-24	-24A						

METHOD OF RATING: IN ACCORDANCE WITH REFERENCED STANDARD.
CAPACITIES: AS PER TABLE I, BASED ON USE OF REFRIGERANT 12, OR AS PER TABLE II, BASED ON USE OF REFRIGERANT 22.
DIMENSIONS: DIMENSIONS ARE IN INCHES.

APPROVED 28 SEPT 62 REVISED (A) 2 JUNE 1969 (B) 9 JULY 1971 (C) 19 JUN 74

P.A. ARMY - GL Other Cust NAVY - YD USAF - 82	TITLE VALVES EXPANSION, THERMOSTATIC REFRIGERANT-12 AND REFRIGERANT-22	MILITARY STANDARD MS 17982
PROCUREMENT SPECIFICATION MIL-V-23450	SUPERSEDES:	SHEET 2 OF 3

DD FORM 1 DEP 67 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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NOTES:

1. DEFINITIONS:
   (A) FACTORY SUPERHEAT SETTING:
   THE "FACTORY SUPERHEAT SETTING" OF A THERMOSTATIC EXPANSION VALVE IS THE FINAL ADJUSTMENT MADE BY THE MANUFACTURER DURING PRODUCTION TESTS.
   (B) SUPERHEAT CHANGE:
   THE "SUPERHEAT CHANGE" (SUPERHEAT GRADIENT) OF A THERMOSTATIC EXPANSION VALVE IS THE DIFFERENCE BETWEEN NO-FLOW SUPERHEAT AND THE SUPERHEAT REQUIRED TO PRODUCE RATED CAPACITY OR REQUIRED FLOW.

2. OPERATING CHARACTERISTICS:
   (A) SUPERHEAT:
   1. TYPE: ADJUSTABLE
   2. SETTING: FOR REFRIGERANT 12 PARTS -1, -2, -3, -4, and -5; FACTORY SET AT 7F ± 2F AT BULB REFERENCE TEMPERATURE OF 0 F.
   PARTS -6 TO -12, INCL.; FACTORY SET AT 10F ± 1F AT BULB REFERENCE TEMPERATURE OF 32F.
   FOR REFRIGERANT 22 PARTS -13, -14, -15, -16 and -17; FACTORY SET AT 6F ± 2F AT BULB REFERENCE TEMPERATURE OF 32F. PARTS -18 TO -24 INCL; FACTORY SET AT 6F ± 1F AT BULB REFERENCE TEMPERATURE OF 32F.
   3. SUPERHEAT CHANGE: THE SUPERHEAT CHANGE OF A THERMOSTATIC EXPANSION VALVE SHALL NOT BE MORE THAN 7F. THROUGHOUT THE RANGE OF TEMPERATURE IN WHICH THE EVAPORATOR IS SPECIFIED TO OPERATE TO PRODUCE THE REQUIRED CAPACITY AT THOSE CONDITIONS.

   (B) PRESSURE LIMIT MEANS: (REFRIGERANT 12 ONLY)
   1. SETTING: PARTS -1 AND -2; FACTORY SET AT MAXIMUM OPERATING PRESSURE OF 30 PSIG., (+) 7 PSI., (-) 0 PSI.

3. MATERIALS: SUITABLE FOR USE WITH REFRIGERANT 12, OR REFRIGERANT 22 AS APPLICABLE.

4. CONNECTION ENDS - FLARE: EXTERNAL, IN ACCORDANCE WITH MS35873.
   SOLDER: FEMALE, IN ACCORDANCE WITH MS35925.

5. MARKING: DIRECTION OF FLOW TO BE LEGIBLY AND PERMANENTLY INDICATED.

6. REFERENCED DOCUMENTS: ANSI STANDARD B 9.1 - SAFETY CODE FOR MECHANICAL REFRIGERATION.
   ARI STANDARD NO. 750-70 - THERMOSTATIC REFRIGERANT EXPANSION VALVES.

7. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS OR REQUEST FOR PROPOSAL.

8. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

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